



**CALICUT UNIVERSITY – FOUR-YEAR  
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

**BSc PHYSICS HONOURS**

|                |                                                                                                                                                                                                                |                  |                   |                    |             |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Programme      | <b>B.Sc. Physics Honours</b>                                                                                                                                                                                   |                  |                   |                    |             |
| Course Title   | <b>ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING IN PHYSICS</b>                                                                                                                                                 |                  |                   |                    |             |
| Type of Course | <b>Major Elective</b>                                                                                                                                                                                          |                  |                   |                    |             |
| Semester       | <b>VIII</b>                                                                                                                                                                                                    |                  |                   |                    |             |
| Academic Level | <b>400 - 499</b>                                                                                                                                                                                               |                  |                   |                    |             |
| Course Details | Credit                                                                                                                                                                                                         | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 4                                                                                                                                                                                                              | 4                | -                 | -                  | 60          |
| Pre-requisites | 1. Fundamental Programming Concepts in Python<br>2. Basic idea of statistics and linear algebra                                                                                                                |                  |                   |                    |             |
| Course Summary | This course explores the fundamentals of Artificial Intelligence and Introduces the basic concepts of Machine Learning Techniques. Also explores various clustering, classification and regression techniques. |                  |                   |                    |             |

**Course Outcomes (CO):**

| CO                                                                                                                                                                                                | CO Statement                                                                                                                                                                     | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|--------------------------------------------------------|
| CO1                                                                                                                                                                                               | Grasp the concepts and importance of Artificial Intelligence, historical context and how the brain processes information.                                                        | U                | C                   | Instructor-created exams / Quiz                        |
| CO2                                                                                                                                                                                               | Acquire a solid understanding of machine learning principles, algorithms, and evaluation techniques and apply them effectively to real-world problems.                           | U                | C                   | Instructor-created exams / Home Assignments            |
| CO3                                                                                                                                                                                               | Understand neural networks, perceptron, linear regression, and multilayer perceptron (MLP) and practical implementation for real-world problems using MLP.                       | Ap               | P                   | Seminar Presentation / Group Tutorial Work             |
| CO4                                                                                                                                                                                               | Acquire a comprehensive understanding of deep learning models, their comparison with traditional machine learning, various types of deep neural networks and their architecture. | U                | C                   | Instructor-created exams / Home Assignments            |
| CO5                                                                                                                                                                                               | Design and develop machine learning models using Keras and MLP for various problems in the real world.                                                                           | Ap               | P                   | Practical Assignment / Observation of Practical Skills |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)<br>Metacognitive Knowledge (M) |                                                                                                                                                                                  |                  |                     |                                                        |

**Detailed Syllabus:**

| <b>Module</b> | <b>Unit</b>                                    | <b>Content</b>                                                                                                                                                  | <b>Hrs<br/>(48<br/>+12)</b> | <b>Marks<br/>(70)</b> |
|---------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------|
| <b>I</b>      | <b>FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE</b> |                                                                                                                                                                 | <b>10</b>                   | <b>15</b>             |
|               | 1                                              | What is Artificial Intelligence - Turing Test - Cognitive modeling approach                                                                                     | 2                           |                       |
|               | 2                                              | Foundations of AI - How do brain process information - How can we build an efficient computer                                                                   | 3                           |                       |
|               | 3                                              | History of AI - The birth - Early Enthusiasm - Knowledge-based systems - AI adopts the scientific method - Intelligent agents - Availability of large data sets | 4                           |                       |
|               | 4                                              | The State of art                                                                                                                                                | 1                           |                       |
|               | Section 1.1 - 1.4 of Book 1                    |                                                                                                                                                                 |                             |                       |
| <b>II</b>     | <b>FOUNDATIONS OF ML</b>                       |                                                                                                                                                                 | <b>12</b>                   | <b>18</b>             |
|               | 5                                              | Introduction to Machine Learning - Learning - Types of Machine Learning                                                                                         | 1                           |                       |
|               | 8                                              | Supervised Learning - Regression - Classification                                                                                                               | 2                           |                       |
|               | 9                                              | Learning process - Terminology - Weight Space                                                                                                                   | 3                           |                       |
|               | 10                                             | Testing machine learning algorithms - Training - Testing - Validation - Matrices to evaluate the model                                                          | 3                           |                       |
|               | 11                                             | Turning data into probabilities - Basic statistics - Bias Variance tradeoff                                                                                     | 3                           |                       |
|               | Chapter 1 & 2 of Book 2                        |                                                                                                                                                                 |                             |                       |
| <b>III</b>    | <b>ARTIFICIAL NEURAL NETWORKS</b>              |                                                                                                                                                                 | <b>17</b>                   | <b>25</b>             |

|           |                                   |                                                                                                                                                                                                                                |           |           |
|-----------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
|           | 12                                | The brain and neuron - McCulloch and Pitts Neurons - Neural Networks                                                                                                                                                           | 2         |           |
|           | 13                                | Perceptron - Bias - Learning rate - Perceptron learning algorithm - implementation                                                                                                                                             | 3         |           |
|           | 14                                | Linear Separability - Perceptron Convergence Theorem                                                                                                                                                                           | 3         |           |
|           | 15                                | Linear Regression - An example problem                                                                                                                                                                                         | 2         |           |
|           | 16                                | Multi-layer Perceptron -Forward Network - Backpropagation of Errors - Algorithmic details                                                                                                                                      | 4         |           |
|           | 17                                | How to implement MLP - Data - Training - Overfitting                                                                                                                                                                           | 2         |           |
|           | 18                                | Overview of different problems using MLP - Steps involved in MLP                                                                                                                                                               | 1         |           |
|           | Chapter 3 & 4 of Book 3           |                                                                                                                                                                                                                                |           |           |
| <b>IV</b> | <b>DEEP LEARNING FUNDAMENTALS</b> |                                                                                                                                                                                                                                | <b>9</b>  | <b>12</b> |
|           | 19                                | Deep Learning - Working of DL Model - Comparison between DL and ML                                                                                                                                                             | 1         |           |
|           | 20                                | Applications of Deep Learning - Libraries for implementing DL - TensorFlow and Keras                                                                                                                                           | 2         |           |
|           | 21                                | Types of Neural Networks - ANN - MLP - CNN - RNN                                                                                                                                                                               | 4         |           |
|           | 22                                | Architecture of Keras - Model - Layer - Loss - Optimizer - Metrics                                                                                                                                                             | 2         |           |
|           | Section 12.1 - 12.4 of Book 3     |                                                                                                                                                                                                                                |           |           |
| <b>V</b>  | <b>MINI PROJECT: OPEN ENDED</b>   |                                                                                                                                                                                                                                | <b>30</b> |           |
|           | 1                                 | Implement the following:<br><br><b>1. Design a MLModel:</b> With ionosphere data to identify any structure is present in a radar data using Keras<br><br><b>2. Design ML Classifier:</b> To classify RR Lyrae stars using KNN. |           |           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                      |                                                                                                                                                         |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                      | <b>3. Design a MLP Classifier for classification problems:</b> Data can be anything including the topics in Physics, Astrophysics, Climate Studies etc. |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sections from References:<br><br>1. Section 12.4 of Book 3<br><br>2. <a href="https://sigmoidal.ai/en/k-nearest-neighbors-k-nn-for-classifying-rr-lyrae-stars/">https://sigmoidal.ai/en/k-nearest-neighbors-k-nn-for-classifying-rr-lyrae-stars/</a> |                                                                                                                                                         |  |  |
| Books and References:<br><br>1. Artificial Intelligence – A Modern Approach Third Edition by Stuart Russel and Peter Norvig. Section 1.1 - 1.4 (Book 1)<br><br>2. Machine Learning: An Algorithmic Perspective by Stephen Marsland (Book 2)<br><br>3. Data Science and Machine Learning using Python by Reema Thereja (Book 3)<br><br>4. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, Third Edition by Aurélien Géron.<br><br>5. Machine Learning in Data Science using Python by R Nageswara Rao |                                                                                                                                                                                                                                                      |                                                                                                                                                         |  |  |

### Mapping of COs with PSOs and POs :

|      | PSO 1 | PSO 2 | PSO 3 | PSO 4 | PS O5 | PS O6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO 6 | PO 7 |
|------|-------|-------|-------|-------|-------|-------|-----|-----|-----|-----|-----|------|------|
| CO 1 | 2     | 0     | 1     | 0     | 2     | 1     | 3   | 1   | 1   | 1   | 2   | 1    | 1    |
| CO 2 | 2     | 1     | 2     | 0     | 2     | 1     | 3   | 2   | 1   | 2   | 3   | 1    | 1    |
| CO 3 | 1     | 2     | 3     | 1     | 2     | 2     | 3   | 1   | 1   | 2   | 3   | 2    | 1    |
| CO 4 | 2     | 0     | 2     | 0     | 2     | 1     | 3   | 1   | 1   | 2   | 3   | 1    | 1    |
| CO 5 | 1     | 1     | 2     | 2     | 3     | 2     | 3   | 2   | 2   | 3   | 3   | 2    | 1    |

### Correlation Levels:

| Level | Correlation        |
|-------|--------------------|
| 0     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

**Mapping of COs to Assessment Rubrics**

|      | Internal Theory/<br>Practical Exam | Assignment<br>/Viva | Practical Skill<br>Evaluation | End Semester<br>Examinations |
|------|------------------------------------|---------------------|-------------------------------|------------------------------|
| CO 1 | ✓                                  | ✓                   |                               | ✓                            |
| CO 2 | ✓                                  | ✓                   |                               | ✓                            |
| CO 3 | ✓                                  | ✓                   |                               | ✓                            |
| CO 4 | ✓                                  | ✓                   |                               | ✓                            |
| CO 5 |                                    | ✓                   | ✓                             |                              |